Involutions outside the reversal and subgroup obstructions

Human Compression project

Theorem 1. *Let G be a finite simple exceptional group of Lie type, other than $G_2(2)' \cong \mathrm{PSU}_3(3)$ and ${}^2G_2(3)' \cong \mathrm{PSL}_2(8)$. There are $x, t \in G$ such that*

$$|t| = 2, \qquad \langle x, t\rangle = G, \qquad \nexists \alpha \in \mathrm{Aut}(G) : \ (x^\alpha, t^\alpha) = (x^{-1}, t).$$

This is the exceptional-group part of the positive existence theorem of Leemans and Liebeck [1]. The already reviewed finite certificates cover $G_2(3)$, $G_2(4)$ and the Tits group ${}^2F_4(2)'$. The argument below treats every remaining parameter. The passage from these generators to an abstract chiral polyhedron is the retained Schulte–Weiss reconstruction theorem. In a nonabelian simple group the cyclic-subgroup intersection required there is automatic: $\langle x\rangle \cap \langle x^{-1}t\rangle$ centralizes both generators, and is therefore trivial.

1 A counting criterion with the outer automorphisms included

For a finite group H, write $i_2(H)$ for its number of elements of order two and $\nu_2(H) = i_2(H) + 1$. Thus ν_2 counts solutions of $h^2 = 1$, including the identity. If $K \lhd H$, then

$$\nu_2(H) \le |K|\nu_2(H/K). \tag{1}$$

Indeed, each solution in H maps to a solution in the quotient, and a fibre contains at most $|K|$ elements. If K is central of odd order, each such fibre contains at most one solution: any two differ by a central element whose square is one. We will use both forms, keeping the identity in every estimate involving an extension.

Lemma 2. *Let G be centreless, let $G \le J \lhd A = \mathrm{Aut}(G)$, and suppose $|x| > 2$ and $C_J(x) = T$. Put*

$$\kappa = |\{aJ \in A/J : (aJ)^2 = J\}|.$$

Suppose μ bounds $i_2(C_G(\alpha))$ for every involution $\alpha \in A$ which sends x to x^{-1}. Suppose every proper subgroup containing x lies in either $N_G(\langle x\rangle)$ or one of the listed maximal subgroups $M_1, \dots, M_s$. If

$$i_2(G) > \kappa|T|\mu + |N_G(\langle x\rangle)| + \sum_{j=1}^{s} i_2(M_j), \tag{2}$$

then x has an involutory partner as in Theorem 1.

Proof. Fix a coset of J containing a reverser a. Every other reverser in that coset differs from a by an element of $C_J(x)$; this follows by applying their quotient to x. The set of reversers in that coset thus has at most $|T|$ elements. An involutory reverser has image of square one in A/J, so there are at most $\kappa|T|$ of them in total. The first term of (2) consequently bounds the involutions fixed by any involutory reverser. The remaining terms bound all involutions which fail to generate G with x. Choose t outside this union. If an automorphism reverses x and fixes t, its square fixes the generating pair and is the identity. It is not the identity, because $|x| > 2$. It would therefore be one of the excluded involutory reversers, a contradiction. □

The source uses the corresponding single-coset estimate. Our form only requires the centralizer inside J; field and graph automorphisms are counted explicitly. It also charges the torus normalizer explicitly.

2 The tori and the retained subgroup classification

Let $\mathbf{G}$ be the adjoint algebraic group, let F be its defining endomorphism, and put $J = \mathbf{G}^F$, $G = J'$. Choose the cyclic torus $T_0 = \langle x\rangle$ in Table 1. Its existence and cyclicity are the torus results collected in

G	D	other maximal overgroups of T_0
$E_8(q)$	$\Phi_{30}(q)$	none
$E_7(q),\ q \geq 3$	$\Phi_{18}(q)(q+1)$	one ${}^2E_6(q).(q+1)/d$
$E_7(2)$	129	one $\mathrm{PSU}_8(2).2$
$E_6(q)$	$\Phi_9(q)$	one $\mathrm{PSL}_3(q^3).3$
${}^2E_6(q)$	$\Phi_{18}(q)$	one $\mathrm{PSU}_3(q^3).3$
$F_4(q),\ q \geq 3$	$\Phi_{12}(q)$	at most two ${}^3D_4(q).3$
$F_4(2)$	17	two $\mathrm{Sp}_8(2)$
$G_2(q),\ q \geq 5$	$\Phi_6(q)$	one $\mathrm{SU}_3(q).2$
${}^3D_4(q)$	$\Phi_{12}(q)$	none
${}^2F_4(q),\ q \geq 8$	$q^2 + \sqrt{2q^3} + q + \sqrt{2q} + 1$	none
${}^2G_2(q),\ q \geq 27$	$q + \sqrt{3q} + 1$	none
${}^2B_2(q),\ q \geq 8$	$q + \sqrt{2q} + 1$	none

Table 1: The chosen tori and their other maximal overgroups.

Kantor–Seress [3, Section 2]. The index of T_0 in the corresponding torus T of J is $d = (2, q-1)$ for E_7, $d = (3, q-\epsilon)$ for E_6^ϵ, and $d = 1$ otherwise. In the table D is the exact order of T.

The overgroup information, including the numbers of overgroups containing the chosen torus rather than just their conjugacy types, is retained from Guralnick–Kantor [2, Propositions 6.1–6.2 and their proofs]. Their small-case Table IV supplies $F_4(2)$ and $E_7(2)$. Central quotients in Table 1 are understood in their natural embeddings; the estimates below allow a covering kernel wherever needed. We are not reproving the classification of maximal subgroups.

We do prove the centralizer interface required for Lemma 2. This avoids applying a general torus-centralizer statement outside its stated field-size range.

Lemma 3. *For the choices in Table 1, $C_{\mathbf{G}}(x) = \mathbf{T}$ is the connected algebraic torus containing T, and $C_J(x) = T$. Moreover,*

$$|N_G(\langle x \rangle)| \leq |W|D, \tag{3}$$

where W is the absolute Weyl group.

Proof. Let Q be the character lattice of the adjoint torus, which is the root lattice. For an ordinary twisted or untwisted group the endomorphism on this lattice is $F^* = qu$, with u an isometry. A root α trivial on T_0 gives a character of T/T_0, and hence

$$d\alpha = (qu - I)v \quad \text{for some } v \in Q.$$

Consequently

$$\|v\| \leq \frac{d\|\alpha\|}{q-1}. \tag{4}$$

Equality requires $uv = v$. In the simply laced cases all roots have the length of a shortest nonzero lattice vector. The chosen torus types have no eigenvalue one: their characteristic polynomials are Φ_{30}, $\Phi_{18}\Phi_2$ (or $\Phi_{14}\Phi_2$ for $E_7(2)$), Φ_9, Φ_{18} and Φ_{12}, respectively. For $E_8, E_7, E_6^\epsilon, {}^3D_4$ the right side of (4) is at most this minimum length, except for ${}^2E_6(2)$. If equality could occur, the absence of a fixed vector makes the actual inequality strict. Thus no root is trivial on T_0. For $F_4(q)$ with $q \geq 3$ and $G_2(q)$ with $q \geq 5$, the ratios of longest to shortest root length are $\sqrt{2}$ and $\sqrt{3}$, so (4) is again strictly below the minimum lattice length.

For the Suzuki and Ree endomorphisms, $F^*/\sqrt{q}$ is an isometry. Replacing $q-1$ by $\sqrt{q}-1$ proves the same assertion: at the smallest allowed parameters the relevant upper ratios are $\sqrt{2}/(\sqrt{8}-1) < 1$ and $\sqrt{3}/(\sqrt{27}-1) < 1$. This includes ${}^2F_4(8)$.

The two remaining lattice checks are finite. We use simple-root bases with the E_6 diagram edges $(1,2), (2,3), (3,4), (4,5), (3,6)$ and the F_4 Cartan matrix

$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -2 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}.$$

The supplied certificate constructs all 72 and 48 roots by their simple reflections. It supplies a Weyl matrix w of characteristic polynomial Φ_9 in E_6 and one of polynomial Φ_8 in F_4. For every root, respectively,

$$(-2w - I)^{-1}(3\alpha) \notin \mathbb{Z}^6, \qquad (2w - I)^{-1}\alpha \notin \mathbb{Z}^4.$$

The determinants have absolute values 57 and 17. These are the required two torus types, and the displayed tests prove regularity in ${}^2E_6(2)$ and $F_4(2)$. Thus in all cases $C_{\mathbf{G}}(x)^\circ = \mathbf{T}$.

We check connectedness as well. For a central isogeny $\widetilde{\mathbf{G}} \to \mathbf{G}$ from the simply connected group, the component group of a semisimple centralizer embeds in the central kernel: an element centralizing x takes a lift $\tilde{x}$ to $z\tilde{x}$, and the kernel of this map is connected. If $|x|$ is coprime to the order of the kernel, then $z^{|x|} = 1$ forces $z = 1$. This applies to E_6^ϵ because $\Phi_9(q)/d$ and $\Phi_{18}(q)/d$ are prime to three; when three divides the numerator its valuation is exactly one. It applies to ${}^3D_4$ because its torus has odd order and the covering kernel is a two-group. The other cases have trivial central kernel on geometric points, except for odd-characteristic E_7.

For E_7, the exact Weyl calculation gives $C_W(u) = \langle u \rangle$ of order 18, or 14 at $q = 2$. In the rational normalizer u acts on T by the inverse of the q-power map. For $q \geq 3$, $\Phi_{18}(q)$ divides $|T_0|$, and q has order 18 modulo $\Phi_{18}(q)$: its order divides 18, $q^9 \equiv -1$, and $q^6 \equiv q^3 - 1 \not\equiv 1$ modulo that number. These exclude every proper divisor of 18. At $q = 2$, the order of two modulo 129 is 14. Thus no nonidentity Weyl-normalizer element fixes x. If its algebraic centralizer were disconnected, its component group would have order two. Frobenius acts trivially on that group, and Lang's theorem for the connected torus would give a rational representative of its nontrivial component. That contradicts the faithful normalizer action. The centralizer is therefore connected here too. The certificate checks both Weyl centralizers, not merely the orders of chosen elements.

Finally, a normalizer of $\langle x \rangle$ normalizes its algebraic centralizer $\mathbf{T}$. Its quotient by its intersection with $\mathbf{T}$ embeds in W, while that intersection has at most D elements. This proves (3). □

3 Involutions which reverse the torus generator

Write $q = p^f$. The standard structure of outer automorphisms gives the following bounds on the square-one elements of $\mathrm{Aut}(G)/J$:

G	E_8	E_7	E_6^ϵ	F_4	G_2	${}^3D_4$	${}^2F_4$	${}^2G_2$	${}^2B_2$
κ	2	2	4	2	2	2	1	1	1.

The quotient is cyclic apart from the untwisted E_6 graph factor, where it is $C_f \times C_2$. In the exceptional graph-field cases F_4 in characteristic two and G_2 in characteristic three it is cyclic of order $2f$. For ${}^3D_4$ it is cyclic of order $3f$, so $\kappa = 1$ at $q = 2, 3$. In the three Suzuki–Ree families the field degree is odd and there is no nontrivial element of square one.

Suppose first that an inner-diagonal involution reverses x. For every type other than E_6, the longest Weyl element is -1. An algebraic representative n of it can be chosen to have order two; all elements in $\mathbf{T}n$ then have order two and are conjugate over the algebraic closure, since squaring on the torus is surjective. There is a rational representative by Lang's theorem for $\mathbf{T}$. If a is any other rational reverser, an^{-1} centralizes x and hence lies in T, by Lemma 3. Thus all such reversers belong to this algebraic involution class.

For E_6 the Weyl centralizer of the Φ_9 torus element has order nine, as checked in the root certificate. The twisted torus uses its negative; its Weyl centralizer is the same group. Since the torus order is odd, $N_J(T)$ has odd order. There is consequently no inner-diagonal involutory reverser in this case.

The involution centralizer classification is a retained input here. For odd characteristic it is summarized in [5, Section 2, Table 5]; for even characteristic we use Aschbacher–Seitz [6, Sections 13–18] with their corrections [7]. Applied to the lift of -1, it gives the centralizer types in Table 2. For a covering group or an adjoint group the indicated small central kernels are allowed. The powers in brackets denote the order of the normal unipotent subgroup.

To estimate their involutions we retain the uniform theorem of Lawther–Liebeck–Seitz [4, Proposition 1.3]:

$$i_2(\mathrm{Aut}(H)) < 2(q^{M_H} + q^{M_H - 1}). \tag{5}$$

Here $M_H = N + r$ for the absolute root system, and half that value for a Suzuki–Ree group. Since the upper bound is an integer, the same expression is an upper bound for $\nu_2(\mathrm{Aut}(H))$. This observation is needed in using (1).

For E_8 in odd characteristic a central kernel of order at most four reduces the centralizer to a subgroup of $\mathrm{Aut}(P\Omega_{16}^+(q))$. Thus $\nu_2 \leq 8(q^{64} + q^{63})$, which is below $2q^{84}(q^{20} + q^{19})$. For odd E_7, a central kernel of order at most eight gives $\nu_2 \leq 16(q^{35} + q^{34})$, below $2q^{42}(q^{12} + q^{11})$. In even characteristic (1) gives the latter two expressions directly.

type	odd characteristic	even characteristic
E_8	D_8	$[q^{84}].\,\mathrm{Sp}_8(q)$
E_7	$A_7^{\epsilon}.2$	$[q^{42}].\,\mathrm{Sp}_6(q)$
F_4	$(A_1C_3).2$	$[q^{18}].\,\mathrm{SL}_2(q)^2$
G_2	$(A_1A_1).2$	$[q^3].\,\mathrm{SL}_2(q)$
${}^3D_4$	$(A_1(q)A_1(q^3)).2$	$[q^9].\,\mathrm{SL}_2(q)$
${}^2F_4$	—	$[q^9].\,\mathrm{SL}_2(q)$
${}^2G_2$	$\mathrm{PSL}_2(q)\times 2$	—
${}^2B_2$	—	a Sylow two-subgroup

Table 2: Retained centralizer types for a lift of the inverting Weyl element.

For odd q, $\nu_2(\mathrm{PGL}_2(q)) = q^2+1$. The central products in the next three rows have kernel at most two, and their quotients embed in the products of the corresponding inner-diagonal factors. They give

$$\begin{array}{c|c} F_4 & 4(q^2+1)(q^{12}+q^{11}) < q^{22} \\ G_2 & 2(q^2+1)^2 < q^5 \\ {}^3D_4 & 2(q^2+1)(q^6+1) < q^{11}. \end{array}$$

These inequalities hold for odd $q \geq 3$. In even characteristic $\nu_2(\mathrm{SL}_2(q)) = q^2$; multiplication by the indicated unipotent orders gives q^{22}, q^5, q^{11}, and also q^{11} for ${}^2F_4$. For the small Ree group, $i_2(\mathrm{PSL}_2(q)\times 2) = q(q-1)+1 < 2q^2$. For the Suzuki group the acquired Sylow-subgroup calculation gives $i_2(C_G(n)) = q-1$.

The outer involutions must still be included. Proposition 1.1 of [4] gives their fixed groups. A field involution has a group of the same type over s with $s^2 = q$, allowing inner-diagonal extensions. Formula (5) bounds its involutions by $4q^{M_G/2}$. This fits the preceding bounds for $E_8, E_7, F_4, G_2, {}^3D_4$. For E_6^{ϵ} a field or graph-field involution is bounded by $4q^{21}$. A graph involution has fixed group $F_4(q)$, $C_4(q)$, or an involution centralizer in $F_4(q)$; in every case $2(q^{28}+q^{27})$ is sufficient. The exceptional graph-field involutions of F_4 and G_2 have fixed groups ${}^2F_4(q)$ and ${}^2G_2(q)$ respectively. Their bounds $2(q^{14}+q^{13})$ and $2(q^4+q^3)$ fit q^{22} and q^5. At $F_4(2)$ the fixed group is ${}^2F_4(2)$, which embeds by conjugation in the automorphism group of its simple derived Tits group; Proposition 1.3 of [4] still gives the bound $49152 < 2^{22}$. No field involution occurs in the three Suzuki–Ree families themselves.

We have therefore proved that the quantities C in Table 4 bound $i_2(C_G(\alpha))$ for *every* involutory reverser, not only the inner-diagonal ones.

4 Lower bounds from one explicit conjugacy class

We also make the constants in the lower bounds explicit. Let J again denote the inner-diagonal group. Proposition 2.3, Remark 2.2 and Table 5 of [5] provide the involutions in Table 3. Each lies in G, and its J-class is its G-class. Thus $|J|/|C_J(t)|$ is a lower bound for $i_2(G)$. These are retained class orders, not estimates inferred from asymptotic notation.

type	$\lvert C_J(t)\rvert$, odd characteristic	$\lvert C_J(t)\rvert$, even characteristic
E_8	$\lvert\mathrm{SO}_{16}^+(q)\rvert$	$q^{84}\lvert\mathrm{Sp}_8(q)\rvert$
E_7	$2\lvert\mathrm{SL}_8^{\epsilon}(q)\rvert$, $q\equiv\epsilon \pmod 4$	$q^{42}\lvert\mathrm{Sp}_6(q)\rvert$
E_6^{ϵ}	$\lvert\mathrm{SL}_2(q)\rvert\lvert\mathrm{SL}_6^{\epsilon}(q)\rvert$	$q^{27}\lvert\mathrm{SL}_2(q)\rvert\lvert\mathrm{SL}_3^{\epsilon}(q)\rvert$
F_4	$\lvert\mathrm{SL}_2(q)\rvert\lvert\mathrm{Sp}_6(q)\rvert$	$q^{18}\lvert\mathrm{SL}_2(q)\rvert^2$
G_2	$\lvert\mathrm{SL}_2(q)\rvert^2$	$q^3\lvert\mathrm{SL}_2(q)\rvert$
${}^3D_4$	$\lvert\mathrm{SL}_2(q)\rvert\lvert\mathrm{SL}_2(q^3)\rvert$	$q^9\lvert\mathrm{SL}_2(q)\rvert$
${}^2F_4$	—	$q^9\lvert\mathrm{SL}_2(q)\rvert$
${}^2G_2$	$\lvert\mathrm{SL}_2(q)\rvert$	—
${}^2B_2$	—	q^2

Table 3: Centralizer orders of the selected lower-bound classes.

Substitution of the standard group-order products gives

$$i_2(G) > q^a/c, \qquad \begin{array}{c|cccccccccc} G & E_8 & E_7 & E_6^{\epsilon} & F_4 & G_2 & {}^3D_4 & {}^2F_4 & {}^2G_2 & {}^2B_2 \\ \hline a & 128 & 70 & 40 & 28 & 8 & 16 & 14 & 4 & 3 \\ c & 2 & 4 & 2 & 2 & 2 & 2 & 2 & 2 & 2. \end{array} \tag{6}$$

Here is an exact way to check every constant. Form the integer polynomial $P(q) = c|J| - q^a|C_J(t)|$ from the row in Table 3. Expand $P(z + q_0)$, where q_0 is the smallest allowed parameter of that parity. In all eighteen cases every coefficient is nonnegative and the constant coefficient is positive. The certificate stores every coefficient, using integer arithmetic; this proves positivity for all real $q \geq q_0$ and hence for every required prime power. The order products used are

$$|\operatorname{SL}_n^\epsilon(q)| = q^{n(n-1)/2} \prod_{i=2}^{n} (q^i - \epsilon^i), \quad |\operatorname{Sp}_{2m}(q)| = q^{m^2} \prod_{i=1}^{m} (q^{2i} - 1),$$

and the analogous displayed exceptional order products in the certificate. The correction to the ${}^2E_6$ centralizer in [7] is respected. For the Suzuki groups we will use the stronger exact value $(q-1)(q^2+1)$, obtained from their single involution class.

5 The final inequalities

The column B in Table 4 bounds the sum of the involution counts of the other maximal overgroups in Table 1. For E_7, a central fibre of size at most $q+1$ over $\operatorname{Aut}({}^2E_6(q))$ gives the stated expression. The special $E_7(2)$ overgroup is bounded by $2(2^{35} + 2^{34})$, which is smaller. For E_6^ϵ, the extension of index three contributes no new involutions, and any central covering kernel is odd; apply the strengthened form of (1) and (5) over q^3. For F_4 there are at most two overgroups, each in $\operatorname{Aut}({}^3D_4(q))$. For G_2, the possible central kernel of $\operatorname{SU}_3(q).2$ has order at most three, giving $6(q^5 + q^4)$. These estimates deliberately allow every indicated kernel.

G	C	B	$\lvert W\rvert$
E_8	$2q^{84}(q^{20} + q^{19})$	0	696729600
E_7	$2q^{42}(q^{12} + q^{11})$	$2(q+1)(q^{42} + q^{41})$	2903040
E_6^ϵ	$2(q^{28} + q^{27})$	$2(q^{15} + q^{12})$	51840
$F_4,\ q \geq 3$	q^{22}	$4(q^{16} + q^{15})$	1152
$G_2,\ q \geq 5$	q^5	$6(q^5 + q^4)$	12
${}^3D_4$	q^{11}	0	192
${}^2F_4$	q^{11}	0	1152
${}^2G_2$	$2q^2$	0	12
${}^2B_2$	$q-1$	0	8

Table 4: Bounds charged to the two kinds of obstruction.

For E_6^ϵ we may replace D by the larger $\Phi_9(q)$, and for $E_7(2)$ by $\Phi_{18}(2)\Phi_2(2) = 171$. Let L be the right side of (6), using the exact Suzuki value instead. We verify

$$L - \kappa DC - B - |W|D > 0. \tag{7}$$

For the ordinary groups expand the left side at $q = q_0 + z$, with $q_0 = 2$ for E_8, E_7, E_6^ϵ, $q_0 = 3$ for F_4, $q_0 = 5$ for G_2, and $q_0 = 4$ for ${}^3D_4$. For ${}^2F_4$ and ${}^2B_2$ put $q = 2s^2$ and expand at $s = 2 + z$; for ${}^2G_2$ put $q = 3s^2$ and expand at $s = 3 + z$. Every coefficient is nonnegative and the constant is positive in all nine cases. The supplied rational-arithmetic calculation records the full polynomials and their translated coefficients, rather than sampling field sizes.

At ${}^3D_4(2)$ and ${}^3D_4(3)$ use $\kappa = 1$. The exact margins in (7) are 3648 and 17155227/2. At $F_4(2)$ the selected class has size 350859600; the two overgroups give $B \leq 4(2^{20} + 2^{19})$. With $D = 17$, $C = 2^{22}$ and $\kappa = 2$ the margin is 201942224. These calculations cover every remaining small parameter. Lemma 2, together with (3), now proves Theorem 1.

6 Attribution and verification scope

The existence result belongs to [1]. The choice of tori and the strategy of excluding maximal subgroups and involution centralizers are also borrowed from that proof. The subgroup classification, conjugacy class orders, outer-automorphism classification and uniform upper bound remain explicit external premises, cited where used. They are not new theorems of this project.

The additions here are the outer-coset criterion, the root-lattice centralizer argument including the finite checks, the explicit normalizer charge, and the complete constant inequalities. The latter distinguish

i_2 from ν_2 before passing through an extension. No asymptotic estimate supplies an unverified finite constant. The arithmetic certificates prove precisely the polynomial and lattice assertions they state; they do not prove the retained classification theorems. The full L045 admission additionally requires its classical, alternating and sporadic components. No new conjecture resolution, Lean verification or compression measurement is claimed by this component.

References

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